

Project 2

This project is divided into two parts (Part A and Part B), each with multiple sub-questions. Please submit on Canvas a complete do-file for the entire project, including the requested tables and comments. Carefully review your do-file to ensure it runs without any errors in STATA (meaning that I can use it to replicate your work seamlessly). **Submissions without the do-file will receive a grade of zero.** Submission of the log file is **not** required for this assignment.

Both Part A and Part B will involve constructing certain data variables. For guidance, consult the relevant literature, check online, read some research papers, and construct these variables accordingly. As discussed in class, this assignment includes some steps that may be new to you. The goal is to encourage more independent learning through this exercise. You will be familiar with most of the steps. If some steps are new, I am providing hints and suggestions. Please note that you can write your code and tackle the question you have been asked in many ways. Try to be efficient (a shorter code is better because it minimizes mistakes), sequential (some steps need to be done before other steps to make sure the process runs smoothly), and organized (which is crucial in coding).

The project due date is **Monday, November 24th, at 11:59 PM**. Please, start well in advance and look carefully at the data every time you move from one question to the other. Get back to me should you need any clarifications on the instructions. **However, I will not check the code before submission.**

$$\text{liquidity} = \frac{\text{cash}}{\text{assets}}$$

Part A: Is accumulating cash a good idea?

In the first part of the project, we will analyze the importance of cash reserves in periods of crisis. More specifically, we will focus on the 2007-2008 financial crisis and investigate whether *pre-crisis liquidity levels* can somehow affect the relationship between crises and corporate outcomes. To this aim, let us focus on the following model:

$$(\text{Dependent Variable})_{it} = \alpha + \beta_1 \text{Crisis} + \beta_2 \text{Crisis} * \text{LIQ}_{it2006} + \beta_3 X_{it-1} + \varepsilon. \quad (\text{Model 1})$$

Below is a description of the different items in Model (1):

- We will look at **three** different dependent variables: (1) accounts receivables ratio, (2) short-term debt, and (3) investment.
- The dummy variable **Crisis** takes the value of one during the financial phase of the crisis, specifically from July 1, 2007, to June 30, 2008, and it captures the drop in the supply of bank credit following the onset of the crisis.
- LIQ is a liquidity measure computed at the **end of the second quarter of the year 2006**, one year before the financial crisis.
- X is a set of firm characteristics including (1) firm's size, (2) market to book, (3) sales growth, (4) total debt, (5) tangible assets, (6) net profit margin. Note that variables (1) through (6) are one period lagged (i.e., the X variable has the $t-1$ subscript).

firm → reputation, culture
industry → regulation
year → COVID

i. gvkey
i. sic
i. year

- Model (1) *only* incorporates firm fixed effects, that is, firm time-invariant characteristics (like reputation) that could affect the use of various financing tools.

In Model (1) the liquidity's role is investigated to understand if, and to what extent, it changes or buffers the effects that crises may have on corporate strategies and choices. You will test empirically this relationship.

Question 1 –

- a) Comment on Model (1). Do you think this model is properly accounting for the importance that cash reserves, accumulated right before the crisis, may have on accounts receivables, short-term debt, and investment? Explain why you think this is or is not a good model.
- b) How do you explain that year-fixed effects are not included in Model (1)?
- c) Finally, based on Model (1), it seems that the *LIQ* variable is not included – the interaction between the variable *Crisis* and *LIQ* is included but the independent variable for *LIQ* is not there. Why do you think this is the case?

Question 2 –

Use the dataset **Project2** available in Canvas and construct the variables as described in model 1 (look for the folder DataPS2). After you have constructed these variables, let us “clean” the data and make them ready to run/estimate a few models. To clean the data, follow the strategy suggested by Garcia-Appendini and Montoriol-Garriga (2013) who look at:

“A sample that consists of US-incorporated, non-financial firms from Compustat’s quarterly database with available balance sheet data between the third quarter of 2005 and the fourth quarter of 2010. We exclude firms with non-negative values of total assets (atq), cash (cheq), and sales (saleq), those reporting cash values greater than total assets, and those with missing values of accounts receivable (rectq). As is the standard practice in recent related literature, we also eliminate firms with market capitalization less than \$50 million, as well as firms whose book value of assets is less than \$10 million and those displaying asset or sales growth exceeding 100%.”

Note 1: Compustat is the database we use to retrieve balance sheet and income statement information for US firms. This is exactly where your data are extracted from.

- Winsorize the data and generate summary statistics for these variables. Create a table displaying these summary statistics using the `asdoc` command. Please look up this command if needed, as it enables us to export results to Word, minimizing manual work. Save the “cleaned” version of the data as **Project2_Clean**.

Q3 → tabulate → ESTOUT +
ESTTAB

$\beta_1, \beta_2 \rightarrow \text{sign + magnitude}$
Prof. Francesca Toscano
Quantitative Methods in Finance
Fall 2025
provide rationale

Questions 3 -

Estimate three models and report the results in a table. Refer to this table as *Table 1*. You can tabulate your regression results in a table in Word using a combination of two commands, i.e., `eststo` and `esttab`. I want you to report three columns in *Table 1*. Below are the details:

- Column (1) reports results for model 1 using the accounts receivable ratio as the dependent variable.
- Column (2) reports results for model 1 using the short-term debt as the dependent variable.
- Column (3) reports results for model 1 using investment as the dependent variable.

Include the controls in all the tables. After you have constructed your table, write a brief paragraph discussing the coefficients and the relative magnitude. Spend some time discussing your findings for the coefficient β_2 , that is, our coefficient of interest capturing the impact of the financial crisis for (pre-crisis) cash-rich firms on a variety of corporate dimensions.

Note 2: We have used the `eststo` and `esttab` commands in class. If you missed class, please look up those commands. You need both to generate a table and summarize the results.

Question 4 -

TNIC3HHI → sort
industry year

In this question, you are required to re-estimate Table 1 within two specific sub-groups of firms.

More specifically:

Group 1: It will focus on firms with low market power (i.e., the industry competition they face is strong)

Group 2: It will focus on firms with high market power (i.e., the industry concentration they face is weak).

Data about firm-level exposure to industry concentration can be downloaded at the following link:

<https://hobergphillips.tuck.dartmouth.edu/industryconcen.htm>

This database contains firm-level information on exposure to market concentration. More specifically, it provides a variable, that in the dataset is reported as `TNIC3HHI`, that is positively related to pricing power. So, higher values of `TNIC3HHI` suggest that firms have a high price impact (or power) and the competition they face is weak. For more information on how this variable is generated and its intuition check the following link:

https://hobergphillips.tuck.dartmouth.edu/idata/Readme_tnic3HHIData.txt

To split the sample between low- and high-market power firms, please evaluate if the firm-specific exposure to industry concentration is above or below the yearly industry median. To this aim, use a combination of `sort` and `egen` commands. We have used a similar approach in class.

industry, year
3
compute industry yearly median (egen + by (↓))
compute dummy high - concentration

Q4 → Table

3 for HC = 1 — 6 columns — 3 for HC = 0

After constructing your table, discuss the number of observations. Clearly indicate the initial sample size and whether it decreased after incorporating the industry concentration data. Finally, describe your results and explain what they suggest in relation to your analysis.

Note 3: In Question 4, report results for both groups using six columns in total, three corresponding to Group 1 and three to Group 2. Ensure that your table clearly identifies the group associated with each column.

Part B: Is CSR a cheap talk?

In Part B, I want you to be creative and estimate the following model:

$$(Stock\ Market\ Return)_{it} = \alpha + \beta_1 Crisis + \beta_2 Crisis * \underset{\substack{\downarrow \\ \text{stock price} \\ \text{close price (quarter)}}}{CSR_{it}} + \beta_3 CSR_{it} + \beta_4 X_{it} + \varepsilon$$

(Model 2)

Corporate Social Responsibility (CSR) refers to the idea that companies have responsibilities that extend beyond generating profits. They are expected to contribute positively to society while minimizing their negative impacts on people and the environment. The dataset named CSR, available in the DataPS2 folder, reports firm-level scores on CSR investment. Higher CSR values indicate that a firm has invested more in socially responsible activities, which may, in turn, enhance its reputation.

To compute the stock market return, we will use the simplest method - the simple discrete return. The dataset **Project2** includes quarterly closing prices. Using this information, construct the relative stock market return as the change in the stock price from one period to the next, divided by the previous period's price. Please, always remember to **sort** the data when constructing lag/lead values and changes.

- Perform your analysis yearly, describe your results, and **explain the rationale behind them**. Generate a table and attach to your write-up. As usual, discuss and interpret your results.
- Let's assume you have completed the class and are attending the WSU career fair to explore potential internships or job opportunities. A potential employer, a consulting company, asks about the projects you worked on in your MSF/MBA classes, and you describe the empirical project outlined in Part B of this assignment. After listening, the employer asks:

"Do you think your analysis could be affected by sample bias?"

What does that mean and how would you respond to such question?

Note 4: Stock-market returns are typically volatile and prone to outliers. Don't rely only on mean and standard deviation, but examine median, percentiles, min/max, skewness and kurtosis. Ask yourself whether your results make economic sense.

Discrete: $\frac{P_t - P_{t-1}}{P_{t-1}} \rightarrow \log \rightarrow \text{sort!}$